James Lee

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EDUCATION

University of Toronto **Expected June 2030**

Bachelor of Applied Science (BASc), Mechanical Engineering *Toronto, ON*

- **CGPA:** 3.73/4.00 | Dean's Honour List, 2025–2026
- **Relevant Coursework:** Mechanical Engineering Design, Thermodynamics, Dynamics, Materials Science, Engineering Statistics

PROJECTS

PharmaSlice: Automated Medication Package Separator **Jan. 2026 - Apr. 2026**

Team Lead *University of Toronto*

- Led a 6-member engineering team in designing an automated system for Sinai Health Pharmacy to separate unit-dose medication strips while reducing repetitive hand motions experienced by pharmacy technicians.
- Translated client needs into mechanical design requirements, including 2-, 5-, and 10-package cutting intervals, ±3 mm cutting tolerance, ≥1,750 strips/hour target throughput, and ≥99.9% undamaged-package target reliability.
- Built and programmed an Arduino-controlled prototype using two DC gear motors and a rail-mounted rotary cutting mechanism; implemented automated strip advancement, configurable cutting intervals, start/stop control, and calibration logic. The documented prototype used an Arduino Uno, two 60-RPM DC gear motors, and a rotary trimmer mechanism.
- Produced engineering drawings, assembled and tested the prototype, and was selected as 1 of 16 teams from 200+ projects to present the final design at Myhal.

World Solar Challenge Race Strategy Simulator **Sept. 2026**

Individual Technical Project *Blue Sky Solar Racing Recruitment Project*

- Developed a C++ simulation for the ~3,000 km World Solar Challenge, modelling vehicle energy consumption and generation to determine viable race speeds based on aerodynamic drag, rolling resistance, elevation, electrical losses, and solar irradiance.
- Built on the base constant-speed model with a configurable segment-based optimization strategy that dynamically selects vehicle speed using upcoming elevation, solar irradiance, look-ahead distance, and battery reserve.
- Optimized irradiance lookup by caching the nearest weather-data location, reducing runtime from ~3 seconds per speed iteration to ~4 seconds across the full speed sweep.
- Added whole-race, segment, and per-route-segment strategy modes while keeping given vehicle constants and official race constraints.

WORK EXPERIENCE

Private Swimming Instructor **Aug. 2023 - Aug. 2026**

Self-employed *Markham, ON*

- Built and operated an independent swim-instruction service, sourcing 7 clients through direct outreach and referrals and delivering approximately 100-125 private lessons to students from kindergarten through Grade 8.
- Managed $3,000+ in client revenue, pricing, invoicing, payment tracking, pool reservations, scheduling, and parent communication using Google Sheets, with no safety incidents.
- Adapted individualized coaching for students ranging from first-time to proficient swimmers, helping beginners progress from fear of the water to independently swimming laps and generating 3 referral clients.

LEADERSHIP/ACTIVITIES

Royal Canadian Air Cadet Program **Sept. 2022 - Jun. 2025**

Band Second-in-Command | Marksmanship Team Captain | Assistant Coach *Richmond Hill, ON*

- Led training and administration for a 30+ cadet band, running practices and sectionals, mentoring junior cadets, maintaining attendance/schedules, and coordinating equipment alongside senior leadership.
- Captained and coached a 5-person competitive marksmanship team, analyzing shooting performance during up to 9-12 hours of weekly training and contributing to a 4th-place Ontario provincial team finish.
- Earned 2nd place individually at the Ontario provincial level and helped the marching band achieve 1st place at regional competition.

CERTIFICATIONS, SKILLS & INTERESTS

- **Programming:** Python, C++, MATLAB, Arduino, JavaScript, TypeScript
- **Engineering:** SolidWorks (coursework), mechanical design, engineering drawings, technical analysis
- **Manufacturing:** Manual milling, turning/lathe, drilling, reaming, machine-shop tools
- **Software:** Git/GitHub, Microsoft Excel, Google Sheets
- **Certifications/Training:** George Brown College Basic Machining Course, Standard First Aid & CPR-C
- **Interests:** Olympic-distance triathlon, piano, competitive marksmanship